Year	What	Who	More
1820-1833	Work on the Difference Engine	Charles Babbage	
1833 - Death	Work on the Analytical Engine	Charles Babbage	
1855	Difference Engine is built by 2 swedes	Two Swedes	Took 20 years
1873	Odhner is founded in St. Petersburg		Well respected products, but the company disappeared due to political turmoil. Clones were sold for years afterwards throughout Europe
1876	Tide predicting machine	Lord Kelvin (William Thomson)	
1885	Ferranti is founded in the UK		The company that commercialsed the Manchester Baby. Remained in business into the 1980s
1890	Hollerith comes up with machines to help with the US census	Herman Hollerith	
1912	Profile Tracer	Vannevar Bush	
1920	Atvidaberg Industrie founded in Sweden		Produced the "Facit", a succcesful calculating machine. 1970 they went bankrupt and sold out to Electrolux
1924	Product Integraph	Vannevar Bush	Solved problems in electrical transmission
1928-1931	Differential Analyzer	Vannevar Bush	Solved differential equations
1930	Nautical Almanac has a new director that wants to upgrade the process	Leslie John Comrie	
1931	Bull founded in France		1968 Honeywell took over and they became quite successful within Europe. One of the few companies in Europe that was approaching computing from an administrative perspective
1936	Turing Machine	Alan Mathison Turing	Just a thought experiment, but it embodied all the logical capabilities of a modern computer
1937	Scientific Computing Service Ltd.	Leslie John Comrie	First for-profit computing agency
1938	Konrad Zuse builds a computer in his parets' home		
1939	Atanasoff-Berry Computer	John Vincent Atanasoff & Clifford Berry	Rudimentary electronic computing device
1940	Mechanical computer at Bletchley Park	Alan Mathison Turing	
1941	Zuse KG is founded in Germany	Konrad Zuse	Partly mechanical computing machines. Great for computing. 1967 sold out to Siemens AG
1942	Memorandum "The use of High Speed Vacuum Tube Devices for Calculating"	John Presper Eckert	Ignored by Moore School research dept. director and the army ordnance department
1943	Automatic Sequence Controlled Calculator / Harvard Mk I	Howard Hathaway Aiken / IBM	First tests being run in secret on the Harvark Mk I
1943	"Colossus" Electronic computer at Bletchley Park	Alan Mathison Turing	
1944	Start of Project Whirlwind	MIT Servomechanisms Laboratory	Jay W. Forrester is put in charge of the project
1945	ENIAC completed	Eckert & Mauchly	
1945	IBM founds Watson Scientific Computation Laboratory at Columbia Univ.	IBM	The goal is building an electronic calculator better than the Harvard Mk. I: the SSEC. The SSEC was never intended to be salable
1945	Perry Crawford tells Forrester about digital computers	Perry Crawford	With increased funding they set out to build an electronic computer
1946	Moore School Lectures in the summer		
1946	Mathematisch Centrum established in Amsterdam	Netherlands	

Year	What	Who	More
1947	CPC (Card Programmed Calculator)	IBM	A fast, reliable calculator that showed IBM that there was a market for scientific computing equipment
1948	Selective Sequence Controlled Calculator	IBM	First computer to treat its instructions as data. Had many of the traits of a stored-program computer.
1948	Manchester Baby computer built at Manchester University	Sir Frederic Williams, Max Newman	First stored-program computer
1948	SSEC completed	IBM	It was displayed in the IBM HQ and attracted large crowds but it was already obsolete
1949	EDSAC at Cambridge University	Maurice Wilkes	First practical stored-program computer (program: printing squares of integers
1949	EDVAC completed	Eckert & Mauchly + Von Neumann	First stored-program computer
1949	MDC (Magnetic Drum Computer) & TPM (Tape Processing Machine) under development	IBM	IBM didn't press on with these projects and they didn't hit the market for another 5 years. IBM was focusing on the Defense Calculator
1950	ARCO (Automatic Relais Calculator for Optical problems) build in Delft		
1951	UNIVAC was functional but needed software. EMCC was bought by Remington Rand	Eckert & Mauchly - Remington Rand	Grace Murray hopper was brought on to lead the programming team
1951	Census bureau accepted the UNIVAC	Remington Rand	
1951	project Lincoln (SAGE) starts at MIT, lead by Forrester	MIT	
1951	Bill Papian demostrates Core Memory	MIT	
1951	"The Preparation of Programs for an Electronic Computer" is published by the Cambridge droups		First programming textbook. It set the programming style for the early 50s and the organization of subroutines still follows this proposal to this day
1951	ZERO is created in Delft for the dutch telephone company PTT		
1952	UNIVAC predicts presidential election	remington Rand	
1952	IBM Model 701, 702 announced	IBM	701 = Defense Calculator, 702 = TPM
1952	IBM hires Perry Crawford		
1952	ARRA is presented to the Dutch minister of sciences	Aad van Wijngarden	
1952	Alan Turing designs a chess algorithm		There is no machine to run it on
1953	IBM Model 650 announced	IBM	Based on the MDC, way more popular than the 700 series. More expensive than competitors, but had superior negineering, reliability and software
1953	Core memory in development to replace Williams Tube storage	IBM	
1953	John Backus gets approval to develop FORTRAN for the IBM 704	IBM	
1953	ARRA II is completed		
1953-1956	PETER is built at Philips but not mass-produced because of an agreement with IBM		
1954	Model 704 and Model 705 announced	IBM	updated versions of the IBM model 701 and 702 with core memory
1954	A joint-study team is formed to determine how to best deal with the airline reservations issue	IBM	They recommended a punched card system while the electronic solution (which was still years in the future) was being developed

Year	What	Who	More
1954	First FORTRAN specification is produced	IBM	
1954	the Skinner machine for rote learning		
1955	RAND Corporation is given the contract to write the software for SAGE		
1955	Computer Usage Company (CUC) founded in NY by 2 IBM scientific programmers	Computer Usage Company	First small-time software contractor
1955	ZEBRA is designed in Delft, but built by STANTEC in the UK because Philips was not interested		
1955	Regnecentralen founded in Denmark		Successful range of computers that were sold along with stock in the company. 1989 they were taken over by International Computers Limited (a british company)
1956	ARMAC is completed (successor of the ARRA)		
1956	Electrologica X1 is built based on the ARRA II	Electrologica	
1956	Electrologica is founded with the commercialisation of the ARMAC		Successful in the scientific field but failed to move into administration. Bought by Philips in 1966
1957	Work on the FORTRAN translator is finished	IBM	Westinghouse-Bettin nuclear power facility becomes the first user of FORTRAN
1957	ARPA (Advanced Research Projects Agency) is established as a result of the Sputnik scare		Goal: sponsor and coordinate defense-related research by pursuing projects with long term payoffs, not necessarily immediate results
1957	DEC (Digital Equipment Corporation) is founded	Kenneth Olsen, Harlan Anderson	
1958	Half of the 26 model 704's are using FORTRAN for half their problems	IBM	
1959	IBM 1401 announced	IBM	came from a need to create a transistor-based version of the model 650
1959	FORTRAN II comes out with improvements	IBM	
1959	CUC has 59 employees	Computer Usage Company	
1959	Christopher Strachey proposed the idea of attaching several operating consoles to ainsgle mainframe computer	Christopher Strachey	The idea of time-sharing is born
1960	Specification for SABRE finished and presented to American Airlines	IBM	
1960	COBOL is released	US Government	Heavily influenced by Grace Murray Hopper's FlowMatic
1960	"Man-Computer Symbiosis" is published	J.C.R. Licklider	The manifesto for human computer interaction. He argued that computers should augment our intelect, not rival them. Computer scientists should develop systems that enabled people to enhance their everyday work
1960	DEC PDP-1 released	DEC	125k USD (unbelievable at that time), possible because it was aimed at the science engineering market, not at the data processing one
1960	ALGOL 60		
1960	The game Nimbus is programmed at Regnecentralen		proved that a program could match a human player
1961	Barclays is the first bank to launch a computer center, contracted with De La Rue	Barclays	

Year	What	Who	More
1961	First FORTRAN textbook published	Daniel D. McCracken	
1961	First demonstration of MIT's CTSS (Compatible Time-Sharing System)	Robert Fano, Fernando Corbato	First project of its kind
1961	Spacewar! is created at MIT by 2 students on a PDP-1		
1962	Design began for DTSS (Dartmouth Time-Sharing System)	John Kemeny, Thomas E. Kurtz	
1962	ARPA begins offering lavish funding for Time-Sharing		
1963	MIT's MAC Time-Sharing system went into operation	ARPA / J.C.R. Licklider	Created with 3 million USD funding from ARPA, supported up to 30 users. Could be used to prepare documents, which paved the way for word processors
1963	J.C.R. Licklider set the ARPAnet program into motion	ARPA	
1964	IBM System/360 announced	IBM	
1964	Spectra 70 range of System/360-compatible computers announced	RCA	
1964	Dartmouth BASIC is developed		The goal was to make it so simple that students could start programming after only a few hours of lessons
1965	MIT's MAC is fully operational and highly overloaded		Development started on Multics, but GE was chosen over IBM because they were technologically unsuited for time-sharing
1965	DEC PDP-8 released	DEC	exploited integrated circuits, which made it far smaller and cheaper, at 18k USD
1966	FORTRAN is the first programming language standardized by the American National Standards Institute	IBM	
1966	IBM releases Model 67 as a response to being left out of Multics	IBM	
1966	ARPAnet technology was ready for physical exploitation	ARPA	
1967	De La Rue Automatic Cash System (DACS) put into opreation in North London	Barclays	
1967	CUC has branched out to training, computer services, facilities management, packaged software, consultancy	Computer Usage Company	It had 12 offices, 700 staff memnbers and sales of 13 Million USD annually
1967	Mark IV file management system is released but sales are low	Informatics	After the IBM unbundling, sales skyrocketted
1967	Time-Sharing market is being disputed by 20 companies		
1968	IBM's Unbundling	IBM	
1968	Time-Sharing firms like UCC had terminals across several countries		
1968	Intel incorporates	Intel	
1968	First 4 centers connected to the ARPAnet	ARPA	
1969	Chemical Bank deploys the first US cash dispenser in Long Island, NY		
1969	Bell Laboratories pulls out of Multics		From the Multics debacle came UNIX at Bell Labs
1969	Intel starts developing the 4004 chip	Intel - Ted Hoff	designed for japanese calculator manufacturer Busicom for a scientific calculator

Year	What	Who	More
1970	Competitors of Bank of America's BankAmericard incorporate as National BankAmericard Incorporated (NBI)		
1970	Criticism of Time-Sharing started		
1970	GE withdraws from the Multics project		
1970-1971	First computer recession. RCA and GE give up their computer divisions		
1971	Time-Sharing firms are already in trouble		
1971	Atari is founded	Nolan Bushnell	
1971	The 4004 chip is delivered to Busicom, but the company fails and Intel announces the chip as a "microprogrammable computer on a chip"	Intel	
1971	There are already 23 nodes on the ARPAnet	ARPA	
1972	the Dynabook is conceptualized at Xerox PARC	Alan Kay	
1972	First video game tournament takes place		
1973	Visa's BASE I is delivered	Thompson Ramo Wooldridge (TRW) and McKinsey & Co.	System that provided real-time authorizations for payments
1974	Visa's BASE II is delivered	thompson Ramo Wooldridge (TRW) and McKinsey & Co.	System that handled clearing and settlements
1974	First barcoded products using the UPC start appearing on shelves		
1974	The Intel 8008 replaces the 4004	Intel	Could process 8 bits at a time
1975	Visa introduces Debit Cards named Visa Debit		
1975	Altair 8800 announced	MITS	First computer with a low enough price that it could be afforded by an individual ($397)
1975	Bill Gates and Paul Allen develop a BASIC programming language for the Altair 8800	Bill Gates, Paul Allen	Allen is made director of software at MITS
1975	1,000 email users		The demand for email was a major driving force behind the spread of non-ARPA networks
1975	Homebrew Computer Club in the US		
1976	NBI and Ibanco (their international branch) become Visa and Visa International		
1977	Steve Jobs and Stephen Wozniak launch the Apple II at the West Coast Computer Fair	Apple	
1977	Commodore Business Machines launches the Commodore PET at the West Coast Computer Fair	Commodore Business Machines	
1977	Tandy launches the TRS-80	Tandy	
1977	Hobby Computer Club founded in the Netherlands		
1978	Micro-Pro is founded by Seymour Rubinstein	Micro-Pro	Forst companu to produce successful word-processing software
1978	Micro-Pro releases WordMaster	Micro-Pro	
1978	Minitel is launched in France		

Year	What	Who	More
1979	Micro-Pro releases WordStar	Micro-Pro	
1980	less than 200 hosts on the internet		
1981	IBM launches the IBM PC	IBM	Software made by Microsoft (MS-DOS). The fact that the IBM logo was associated with it legitimized the personal computer and the IBM PC was a huge success
1981	First computer science departments in the netherlands		
1982	GRiD Compass 1101 is launched	GRiD	First laptop computer
1983	DNS is created and standardized	Paul Mockapetris	
1983	Free Software Movement rises from MIT		The idea is that, once purchased, the user is free to do whatever he wants with the piece of software
1984	PC's Limited is founded by Michael Dell	Dell	
1984	Macintosh is released	Apple	Coming in at 2500 USD, initial sales were disappointing. it was relaunched targetting the business market. The macintosh saved apple
1984	only 1000 hosts on the internet		By the late 1980s, most professional computer users had internet access
1984	Information Sciences as an academic programme in the Netherlands at Tillburg		
1985	Dell hires Jay Bell to create a computer based on the intel 80286, called the 286	Dell	
1986	Psion Organizer II PDA is introduced	Psion	An OS-based PDA
1987	30,000 computers have access to the internet		
1988	PC's Limited is renamed to Dell Computer Corporation		
1989	The World Wide Web is proposed at CERN	Tim Berners-Lee, Robert Caillau	It was based on one of Berners-Lee's earlier projects named Enquire
1990	"Archie"	McGill University	combed through the internet and created a directoy of all the downloadable documents online
1990	The Internet is still largely owned by the US Government		
1991	WAIS (Wide Area Information System)	Thinking Machines Corporation of Waltham, Massachussets (TMC)	Searched for documents that matched user-specified keywords
1991	"Gopher"	University of Minnesotta	It was a catalog of catalogs. It let users examine the contents of gopher databases hosted by hundreds of different institutions
1991	Little interest in the Web at conferences		By 1993 it was everywhere
1993	Mosaic Browser	Marc Andreessen	Had a finished feel, like store-bought shrink-wrapped software
1994	Tim Berners-Lee moves to MIT to head the W3C Consortium		
1994	Mosaic Communications Company is incorporated and renamed to Netscape Communications months later'	Marc Andreessen, Jim Clark	
1994	Netscape is made available in december and downloaded by millions	Netscape	
1995	Windows 95 is released	Microsoft	
1995	The Web accounts for a quarter of internet traffic		

Year	What	Who	More
1995	Netscape holds 70% marketshare	Netscape	
1995	NSFnet backbone is shut down	US Government	This ends government ownership of the internet in the US
1995	Amazon.com is launched	Jeff Bezos	
1995	eBay is launched	Pierre Omidyar	
1997	Google is launched	Larry Page, Sergey brin	
1998	Windows 98 is released and Internet Explorer is offered for free	Microsoft	With MS giving IE away for free, it was difficult for Netscape to sell their browser
1998	Google Inc. is incorporated	Larry Page, Sergey brin	
1998	Open Source Movement		Picked up the ideas of the Free Software Movement but interpreted them as free meaning at no cost
1999	Blackberry PDA is launched	RIM	
2001	Google hires professional CEO Eric Schmidt	Google	
2005	Android Inc. is acquired by Google	Google	
2007	The iPhone is launched	Apple	
2012	Minitel ends in France		